Cloudscape Radio: The Ambient Sonification of Clouds

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Poster

The everyday auditory experience for the city dweller is one that is becoming more and more removed from that of a natural setting. The auditory environment of an office, for example, is simple, repetitive and monotonous, and lacking the elements of complexity, variation and flux found in nature. Sonification, however, can be used to collect data from nature and present it aurally, in order to create new acoustic environments as rich as natural ones. Sonifications of nature data can both facilitate a better understanding of nature and provide new aesthetic experiences. Examples of nature data sonification include Garth Paine's *Reeds* (Paine), a sonification of local weather data, Carrie Bodle's Listening Up (Bodle), a sonification of upper atmosphere ions and electrons, and Andrea Polli's *Atmospherics* (Polli), a sonification of storm simulation data.

The complexity, beauty and variety of form that clouds possess inspired the idea of transforming live cloud images into sound – to create a sonic window to the outside world for indoor environments. By installing a live cloud sonification system as an ambient sound system in a place where we carry out our daily affairs, we can maintain a constant awareness of the activity of the natural world which would not be possible visually. Rather than simply eliminating noise pollution and creating barren acoustic environments, we can mask unwanted noises and create acoustic environments that create a sense of place for the individual. (Wrightson, 2000)

It was reasoned that the experience of listening to clouds should be more akin to listening to natural sounds such as wind, rain or seashore sounds, than to listening to conventional music. It was decided that the sound generated should be abstract, non-representational and should not contain any conventional musical pitch, harmony, tempo, or timbre. It was intended that an understanding of the relationship between the various types of clouds and the various types of clouds could be learnt through experience, without the need of iconic or metaphoric representations.

Cloudscape Radio, a system for the sonification of cloud imagery, was developed where images are captured from a USB Webcam and sonified in real-time. Additionally, the capability of sonifying images from publicly available webcams on the internet was built into the software. A large database of Internet webcams, which were pointed at the skies, was collated, so that users can tune into live sonifications of cloudscapes from all over the world, at any time.

A mapping from image data to sound was chosen where the x-axis is mapped to time, the y-axis is mapped to frequency and the single-channel pixel value is mapped to amplitude (Fig. 1), similar to the mapping used in 'the vOICe' image sonifier (Miejer, 1992). However, rather than synthesizing tones consisting of inharmonically related sinusoids, we chose to synthesize 'coloured-noise (noise with a varying spectral envelope) (Hanna, 2004), where each row is mapped to a narrow-band noise signal. Through research into psychoacoustics, we intended to develop an ideal coloured-noise synthesizer that had the capability of perfectly synthesizing any sound within the set of all perceivable coloured-noise sounds.

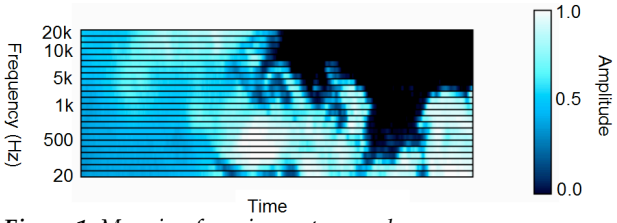


Figure 1. *Mapping from image to sound*

A method for creating a noise signal with an arbitrarily detailed (but static) spectral envelope was developed where a signal containing thousands of sinusoids is pre-rendered using an Inverse-FFT and then looped. By adding a number of narrow-band noise signals generated using this method, coloured-noise with a dynamic spectral envelope can be generated in real-time. Through informal experiments it was discovered that the maximum frequency resolution of hearing for coloured-noise signals is equivalent to the total possible number of concurrent, non-overlapping critical bands, which is 24 according to the Bark scale (Smith, 1999). By controlling the amplitude envelopes of 24 looping, pre-rendered narrowband noise samples, the entire gamut of perceptible coloured-noise signals can be efficiently generated in real-time. By mapping live cloud images to our coloured-noise synthesizer, the complex and dynamic properties of clouds can be preserved to create a rich and constantly evolving acoustic environment.

Installation Requirements

- A PC running Windows XP with the Java Runtime Environment (JRE) 1.4 or greater installed.
- Internet access
- Speakers
- (optional) A USB webcam

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